

# A Literature-Implied Negative Answer to the Asymptotic- $c_0$ Subspace Question

Status packet for arXiv:1607.03587 and arXiv:1902.10092

June 14, 2026

## Source and Original Problem

The source paper is Freeman–Odell–Sari–Zheng, *On spreading sequences and asymptotic structures*, arXiv:1607.03587 [1]. Problem 4.5 asks whether the following hypotheses force either an asymptotic- $c_0$  subspace or a subspace with separable dual:

$\ell_1 \not\hookrightarrow X$  and every weakly-null spreading model in  $X$  is equivalent to  $c_0$ .

$I_n = \{(k_1, k_2, \dots, k_n) : 1 \leq k_1 < \dots < k_n\}$ .

The following question is open.

**Problem 4.5.** Suppose that  $\ell_1$  does not embed into  $X$  and every spreading model generated by weakly null normalized sequences in  $X$  is equivalent to the unit vector basis of  $c_0$ . Does  $X$  contain an asymptotic- $c_0$  subspace? Does  $X$  contain a subspace  $Y$  with  $Y^*$  separable?

Note that the space  $JH$  constructed by Hagler [H] has non separable dual, does not contain  $\ell_1$  and every weakly null normalized sequence has a subsequence equivalent to the unit vector basis of  $c_0$ . So if the problem has affirmative answer it is necessary to pass to a subspace. We will prove a weaker theorem.

**Theorem 4.6.** *Suppose that a Banach space  $X$  does not contain an isomorphic copy of  $\ell_1$  and every asymptotic model  $(e_i)$  generated by weakly null arrays in  $X$  is equivalent to the unit vector basis of  $c_0$ . Then*

This packet records a negative answer only to the first question, the existence of an asymptotic- $c_0$  subspace.

## Supporting Literature

The supporting paper is Argyros–Motakis, *On the complete separation of asymptotic structures in Banach spaces*, arXiv:1902.10092 [2]. Their Theorem A constructs a reflexive Banach space  $X_{\mathbf{iw}}$ .

[FOSZ] states that  $(b)_\infty \Leftrightarrow (a)_\infty$ . This last result requires the coordinate free definition of asymptotic- $\ell_p$  from [MMT].

The problem of Odell that was mentioned earlier in the introduction can be formulated in the language of this paper as follows: is there  $1 \leq p \leq \infty$  so that  $(c)_p \not\approx (a)_p$ ? We actually prove something deeper, namely that  $(c)_p \not\approx (b)_p$  for all  $1 \leq p \leq \infty$ . We also prove  $(d)_1 \not\approx (c)_1$ , although the same argument works for  $1 < p < \infty$  (as it was mentioned earlier  $(c)_\infty \Leftrightarrow (d)_\infty$ ). To achieve these results we present three constructions of Banach spaces. Let us describe the properties of these spaces one by one and later give an outline of how they are defined. The first construction yields  $(c)_1 \not\approx (b)_1$ .

**Theorem A.** There exists a reflexive Banach space  $X_{\mathbf{iw}}$  that has a 1-unconditional basis and the following properties:

- (i) every normalized weakly null sequence in  $X_{\mathbf{iw}}$  has a subsequence that generates a spreading model that is 4-equivalent to the unit vector basis of  $\ell_1$ .
- (ii) every infinite dimensional subspace of  $X_{\mathbf{iw}}$  contains an array of normalized weakly null sequences that generate the unit vector basis of  $c_0$  as an asymptotic model.

Their Theorem C gives the relevant behavior of the dual  $X_{\mathbf{iw}}^*$ .

discrete closed subset  $F$  of  $[1, \infty]$  a space is constructed with stable Krivine set  $F$ . The fact  $(c)_\infty \not\approx (b)_\infty$  is not achieved via a separate construction.

**Theorem C.** The space  $X_{\mathbf{iw}}^*$  has the following properties.

- (i) Every normalized weakly null sequence has a subsequence that generates a spreading model that is 4-equivalent to the unit vector basis of  $c_0$ .
- (ii) Every infinite dimensional subspace of  $X_{\mathbf{iw}}^*$  contains an array of normalized weakly null sequences that generate the unit vector basis of  $\ell_1$  as an asymptotic model.

That is,  $(c)_\infty \not\approx (b)_\infty$  and in particular  $(c)_\infty \not\approx (a)_\infty$ .

We additionally observe that the spaces  $X_{\mathbf{iw}}$  and  $X_{\mathbf{iw}}^*$  are asymptotically symmetric and obtain a negative answer to [JKO, Problem 0.2].

The same paper explicitly states in Section 10 that  $X_{\mathbf{iw}}^*$  has no asymptotic- $c_0$  subspace.

**Remark 9.8.** Because  $X_{\mathbf{iw}}^p$  has a uniformly unique  $\ell_p$ -spreading model the strong Krivine set of every block subspace of  $X_{\mathbf{iw}}^p$  is the singleton  $\{p\}$ .

## 10. THE SPACE $X_{\mathbf{iw}}^*$

In this section we study the space  $X_{\mathbf{iw}}^*$ . We prove that every normalized block sequence in  $X_{\mathbf{iw}}^*$  has a subsequence that generates a spreading model that is 4-equivalent to the unit vector basis of  $c_0$ . In addition, every block subspace of  $X_{\mathbf{iw}}^*$  admits the unit vector basis of  $\ell_1$  as an asymptotic model and hence  $X_{\mathbf{iw}}^*$  does not have an asymptotic- $c_0$  subspace.

**Lemma 10.1.** Let  $j_0 \in \mathbb{N}$ ,  $(g_k)_{k=1}^m$  be an  $\mathcal{S}_{n_{j_0}}$ -admissible sequence in

## Idea of the Proof

The only identification needed beyond the cited construction is a basic feature of spreading models: if a sequence generates a spreading model, then all of its subsequences generate the same spreading

model. Argyros–Motakis prove that every normalized weakly-null sequence in  $X_{\mathbf{iw}}^*$  has a subsequence generating  $c_0$ . Therefore any weakly-null sequence which already generates a spreading model can only generate  $c_0$ . Since  $X_{\mathbf{iw}}^*$  is reflexive and has no asymptotic- $c_0$  subspace, it gives the promised negative answer to the first half of Problem 4.5.

## Theorem

**Theorem 1.** *The first question in [1, Problem 4.5] has a negative answer. More precisely, there is a Banach space  $Z$  such that  $\ell_1$  does not embed into  $Z$ , every spreading model generated by a normalized weakly-null sequence in  $Z$  is equivalent to the unit vector basis of  $c_0$ , and  $Z$  contains no asymptotic- $c_0$  subspace.*

*Proof.* Let  $Z = X_{\mathbf{iw}}^*$ , where  $X_{\mathbf{iw}}$  is the Argyros–Motakis space from [2]. By [2, Theorem A],  $X_{\mathbf{iw}}$  is reflexive. Hence  $Z = X_{\mathbf{iw}}^*$  is reflexive as well. Since every closed subspace of a reflexive space is reflexive and  $\ell_1$  is not reflexive,  $\ell_1$  does not embed into  $Z$ .

We next verify the spreading-model hypothesis from [1, Problem 4.5]. Let  $(z_n)$  be a normalized weakly-null sequence in  $Z$  which generates a spreading model  $(e_n)$ . By [2, Theorem C(i)],  $(z_n)$  has a subsequence  $(z_{n_k})$  generating a spreading model 4-equivalent to the unit vector basis of  $c_0$ . But every subsequence of a sequence that generates a spreading model generates the same spreading model. Thus  $(e_n)$  is equivalent to the unit vector basis of  $c_0$ .

Finally, Argyros–Motakis state in Section 10 of [2] that  $X_{\mathbf{iw}}^*$  does not have an asymptotic- $c_0$  subspace. Therefore  $Z$  satisfies the hypotheses of [1, Problem 4.5] while failing the asymptotic- $c_0$  subspace conclusion.  $\square$

## Scope and Limitations

This is a literature-implied answer, not an original construction from this run. The supporting paper does not appear to state that it is answering [1, Problem 4.5]; the implication is obtained by combining its Theorem C and Section 10 with the standard subsequence property of spreading models.

The second question in [1, Problem 4.5], asking for a subspace  $Y \subset X$  with  $Y^*$  separable, is not answered negatively here. Indeed, the example  $Z = X_{\mathbf{iw}}^*$  is reflexive and comes from a space with a countable basis, so  $Z^* \cong X_{\mathbf{iw}}$  is separable.

The run indexes were searched for arXiv:1607.03587, the source title,  $c_0$  spreading-model keywords, asymptotic- $c_0$ , and arXiv:1902.10092 before promotion. A bounded web search on June 14, 2026 found the separate follow-up arXiv:1611.04443 for the source paper’s Argyros conditional-spreading problem; no existing run packet was found for this implication from arXiv:1902.10092.

## Human Review Recommendation

Check the identification between the source problem’s hypothesis and [2, Theorem C(i)], especially the use of subsequences of sequences that already generate a spreading model. Also check that the supporting paper’s “no asymptotic- $c_0$  subspace” statement is being used with the same coordinate-free meaning intended in [1, Problem 4.5].

## References

- [1] D. Freeman, E. Odell, B. Sari, and B. Zheng, *On spreading sequences and asymptotic structures*, Studia Mathematica 240 (2018), no. 1, 69–91; arXiv:1607.03587.
- [2] S. A. Argyros and P. Motakis, *On the complete separation of asymptotic structures in Banach spaces*, Advances in Mathematics 362 (2020), 106962; arXiv:1902.10092.